

Status	Finished
Started	Wednesday, 15 April 2026, 2:20 PM
Completed	Wednesday, 15 April 2026, 3:47 PM
Duration	1 hour 26 mins
Marks	49.00/50.00
Grade	44.10 out of 45.00 (98%)

Question 1

Correct

Mark 3.00 out of 3.00

Select the most appropriate type of control for each of the following scenarios.

A smart trailer trying to make sure it doesn't bump into the truck that is pulling it.

PID control



A robot arm trying to reach for an object on a cluttered table with fragile glasses all over it.

Optimal control



An autonomous military ship, that wants to avoid an enemy.

Game theory



A robot learning how to sweep the floor.

Reinforcement learning



A drone that needs to patrol the skies above a farm, looking for crop damage.

PID control



A mobile robot that needs to find its way out of a maze.

Reinforcement learning



The controls for a nuclear power plant, where even a slight error could cause a meltdown.

Optimal control



A humanoid robot that needs to learn how to walk.

Reinforcement learning



The correct answer is: A smart trailer trying to make sure it doesn't bump into the truck that is pulling it. → PID control, A robot arm trying to reach for an object on a cluttered table with fragile glasses all over it. → Optimal control, An autonomous military ship, that wants to avoid an enemy. → Game theory, A robot learning how to sweep the floor. → Reinforcement learning, A drone that needs to patrol the skies above a farm, looking for crop damage. → PID control, A mobile robot that needs to find its way out of a maze. → Reinforcement learning, The controls for a nuclear power plant, where even a slight error could cause a meltdown. → Optimal control, A humanoid robot that needs to learn how to walk. → Reinforcement learning

Question **2**

Correct

Mark 12.00 out of 12.00

Consider the following second-order system that we want to control:

$$2\ddot{x}_1 + 4\dot{x}_1 - u = 0$$

with

$$y = \dot{x}_1$$

By introducing a change of variables, rewrite this as a first-order system:

$$\dot{x} = Ax + Bu$$

, where

$$\dot{x} = (\dot{x}_1, \dot{x}_2)^T$$

$$A =$$

$$\begin{pmatrix} 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & -2 \end{pmatrix}$$

$$B^T =$$

$$\begin{pmatrix} 0 & 0.5 \end{pmatrix}$$

Which of the following statements are true?

The system is linear:



The system is time-variant:



The system is controllable:

True

✓ because the rank of the controllability matrix is 2 ✓

The system is observable:

False

✓ because the rank of the observability matrix is 1 ✓

Question 3

Correct

Mark 2.00 out of 2.00

Consider the linear dynamical system below.

$$\begin{aligned}\dot{\mathbf{x}} &= \begin{bmatrix} 0 & 0 & 2 \\ -1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u, \\ \mathbf{y} &= \begin{bmatrix} 0 & -1 & 0 \end{bmatrix} \mathbf{x}\end{aligned}$$

Which of the following statements are true:

Select one:

- ☒ a. It is controllable and observable. ✓
- ☐ b. It is controllable but NOT observable.
- ☐ c. It is NOT controllable but observable.
- ☐ d. It is NEITHER controllable NOR observable.

The correct answer is: It is controllable and observable.

Question **4**

Correct

Mark 7.00 out of 7.00

An autonomous car is travelling in a convoy and needs to maintain a fixed distance of **5 metres** behind the leading vehicle. Every second, it calculates the Euclidean distance from its current position to the leading vehicle, and uses PID control to stabilise around the 5m setpoint.

The distances calculated over the first 4 seconds are:

Time	0	1	2	3
Distance (m)	10	7	3	6

Select the answer with the correct proportional control (e), derivative control (d), and integral control (h) at $t = 3$.

Select one:

- ☒ a. $e=1, d=3, h=6$ ✓
- ☐ b. $e=1, d=3, h=5$
- ☐ c. $e=1, d=-3, h=6$
- ☐ d. $e=3, d=1, h=6$
- ☐ e. $e=1, d=3, h=8$

The correct answer is: $e=1, d=3, h=6$

Question **5**

Partially correct

Mark 11.00 out of 12.00

Consider the second-order system that we want to control:

$$\dot{x}_1 = 4x_2$$

$$\dot{x}_2 = 2u$$

With

$$x_1(0) = 1$$

,

$$x_2(0) = 0$$

, where

$$x_1$$

is the position of the system and

$$u$$

is the control variable, and the cost function is given by

$$J(u) = A(x_1(3) - 4)^2 + B(x_2(3))^2 + \frac{C}{2} \int_0^3 [u(t)]^2 dt$$

By introducing an appropriate Hamiltonian, find the control and state trajectory that minimises the cost function for

$$A = 3, B = 1, C = 1$$

. From this, we have that

$$x_1(t) =$$



$$t^3 +$$



$$t^2 +$$

 $t+$ 

Now assume we set

$$A = 1, B = 0, C = 5$$

.

Which of the following statements are true?

We care more about the velocity than the position of the system:



We would have an arbitrary final velocity:



The final position should be close to 1:



We care more about the integral cost than the terminal cost:



Question **6**

Correct

Mark 2.00 out of 2.00

Consider the following zero-sum matrix game, where the given payoffs are to the column player, who is the maximising player.

$$\begin{pmatrix} 5 & 0 & 3 \\ 2 & 1 & 1 \\ 3 & 2 & 4 \end{pmatrix}$$

Which option represents a Nash equilibrium policy for the column player, where each number is the probability of picking that column?

Select one:

- ☐ a. [0.25, 0, 0.75]
- ☐ b. [0, 0, 1]
- ☐ c. [0, 0.75, 0.25]
- ☐ d. [0, 1, 0]
- ☐ e. [0.75, 0.25, 0]
- ☐ f. [0.5, 0.5, 0]
- ☐ g. [0.5, 0, 0.5]
- ☐ h. [0, 0.5, 0.5]
- ☒ i. [1, 0, 0] ✓

The correct answer is: [1, 0, 0]

Question **7**

Correct

Mark 12.00 out of 12.00

Consider the following matrix game, where the given payoffs are to the column player, who is the maximising player. The row player is minimising, and receives the negative of the shown reward:

$$\begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 3 \\ 2 & 2 & 1 \end{pmatrix}$$

The Nash equilibrium policy for the column player, where each number is the probability of picking that column, is given by

[0.667 ✓ , 0 ✓ , 0.333 ✓]

The payoff of the game is 1.667 ✓ .

Which of the following statements are true?

This game is zero-sum:

True



The game is shown here in extensive form:

False



The game shown here is a discrete game because the payoffs are all integers:

False



The Nash equilibrium is the highest payoff that any player can receive in the game:

False



